

★ PRIMER · UNIT 6

Before the Lecture: What Makes a Line Safe

Unit 6 · Support Vector Machines

Watch this first if dot products and margins are new

Twenty minutes of groundwork — how a line is written with a dot product, what a margin measures, and the trick that lets a straight line curve.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Dot products and margins are here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 6 Sits

The course builds one pipeline; here is this unit's stone

Unit 5

done

Three classifiers and anomalies.

Unit 6

you are here

The widest road between two classes.

Unit 7

next

Trees, forests and boosting.

Unit 5 asked which line. This unit asks which line is safest — a different question.

Why 'Widest' Means 'Safest'

Many lines separate the same two classes. The safest is the one furthest from both of them.

A narrow margin

- The line hugs the data.
- One new point crosses it.
- The verdict flips.

A wide margin

- Room on both sides.
- New points stay put.
- It generalises better.

Only the closest points matter. Those are the support vectors, and the rest are ignored.

The Words the Lecture Assumes

Five terms, and the margin section opens up

Word	What it means	Here
Vector	a list of numbers	one customer's features
Dot product	multiply pairwise, then add	how aligned two vectors are
Hyperplane	a flat divider in any dimension	the line, in more than 2-D
Margin	the gap either side of the line	how much room you have
Support vector	a point touching the margin	the only points that matter

In two dimensions a hyperplane is a line. The word only exists so the maths generalises.

The C Knob, in Three Settings

C decides how much you will tolerate a point on the wrong side

C	The model	Result
small	tolerant of mistakes	wide margin, some points inside
moderate	balanced	the usual choice
large	refuses any mistake	narrow margin, fits the noise

Large C is overfitting under another name. You have now met it three times.

When No Straight Line Works

Some classes cannot be split by any straight line. Lift them into a higher dimension and a flat divider appears.

In two dimensions

- One class rings the other.
- No line can separate them.
- You are stuck.

Lift to three

- Add a squared feature.
- One class rises above.
- A flat plane now splits them.

The kernel trick does this without ever computing the higher-dimensional coordinates.

A Dot Product by Hand

Two vectors, four multiplications, one number

Position	w	x	$w \times x$
1	2	3	6
2	-1	4	-4
3	3	1	3
sum	—	—	5

The dot product is 5. Positive means the vectors broadly point the same way.

From Dot Product to a Verdict

The line is a dot product with a constant added

The rule

- Compute $w \cdot x + b$.
- Positive means one class.
- Negative means the other.

With $b = -4$

- $5 - 4 = 1$, so positive.
- This point is class one.
- Distance from the line: small.

Every linear model in this course is the same dot product with a different loss

Hard Margin Against Soft

Real data has an oddball. What you do with it defines the model

Hard margin

- No point may cross.
- One outlier ruins everything.
- Useless on real data.

Soft margin

- A few may cross, at a cost.
- C sets that cost.
- This is what is actually used.

The oddball is the whole argument. Two philosophies, and only one survives contact.

How the Kernel Trick Works

Three moves, and the third is the one that makes it practical

1 · Want

a higher space

Straight lines fail here, but
would work one dimension up.

2 · Notice

only dots matter

The maths only ever needs dot
products, never the
coordinates.

3 · Substitute

the kernel

A kernel returns that dot
product directly. The lift never
happens.

You get the power of the higher space without paying for it. That is the trick.

Two Kernels You Will Meet

The lecture uses both; RBF is the default in practice

Kernel	Shape it can draw	When to use it
Linear	a straight divider	many features, little data
RBF	closed curved regions	the usual first choice
Polynomial	curved, degree by degree	rarely — hard to tune

RBF has its own knob, gamma. Too large and it draws a bubble around every point.

Notation You Will Meet

Four symbols carry the whole unit

Symbol	Read it as	It stands for
w	w	the weight vector — the line's direction
b	b	the bias — where the line sits
$w \cdot x$	w dot x	the dot product of weights and features
C	C	how harshly a violation is punished

w points across the divider. Its length sets how wide the margin is.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>SVC(kernel='linear', C=1.0)</code>	a straight-line support vector machine
<code>SVC(kernel='rbf', gamma=0.1)</code>	the curved version, and its second knob
<code>model.support_vectors_</code>	the handful of points that actually mattered
<code>OneClassSVM()</code>	the unit's anomaly detector
<code>np.dot(w, x)</code>	the dot product you just did by hand

Look at `support_vectors_`. Seeing how few points decide everything is the lesson.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 7

1

"More support vectors means a better fit"

It usually means the opposite — the model is struggling and leaning on half the data.

2

"A large C is the safe setting"

Large C refuses to tolerate any error, so it fits the noise. It is overfitting with a dial.

3

"The kernel actually moves data to a higher dimension"

It never does. It returns the dot product that lift would have produced, and skips the lift.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1 $w = (2, -1, 3)$, $x = (3, 4, 1)$. What is $w \cdot x$?

5

2 Which points determine an SVM's line?

support vectors

3 C is set very large. What is the danger?

overfitting

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It derives the widest road

From 'which line is best' to the margin as a formula, worked by hand.

2

It introduces the C knob

Hard against soft margin, and what one oddball does to each philosophy.

3

It ends in code

The problem set finds the widest road, turns C, and runs a one-class SVM on outliers.

Next: Unit 6 lecture — Support Vector Machines